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SIMATS ENGINEERING

ASSESSMENT TOOL 1

APPLICATION BASED QUESTIONS

1. A factory has three boxes containing colored balls. Box A contains 6 red, 4 blueand 2 green balls; Box B contains 5 red, 7 blue, and 3 green balls; and Box C contains 8 red, 2 blue, and 5 green balls. One box is selected at random, but Box B is twice as likely to be chosen as each of the other boxes. A ball drawn from the chosen box is found to be blue. What is the probability that the selected box was Box C?

Sol)

Given:

- Box A: 6 Red, 4 Blue, 2 Green (12 balls)
- Box B: 5 Red, 7 Blue, 3 Green (15 balls)
- Box C: 8 Red, 2 Blue, 5 Green (15 balls)
- Box B is twice as likely to be chosen.

Let:

- $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$, $P(C) = \frac{1}{4}$

Probability of drawing a blue ball:

- $P(\text{Blue} \mid A) = \frac{4}{12} = \frac{1}{3}$
- $P(\text{Blue} \mid B) = \frac{7}{15}$
- $P(\text{Blue} \mid C) = \frac{2}{15}$

Formula (Bayes' Theorem):

$$P(\boldsymbol{C} \mid \textbf{Blue}) = \frac{\boldsymbol{P}(\boldsymbol{C}) \times \boldsymbol{P}(\textbf{Blue} \mid \boldsymbol{C})}{\boldsymbol{P}(\boldsymbol{A})\boldsymbol{P}(\textbf{Blue} \mid \boldsymbol{A}) + \boldsymbol{P}(\boldsymbol{B})\boldsymbol{P}(\textbf{Blue} \mid \boldsymbol{B}) + \boldsymbol{P}(\boldsymbol{C})\boldsymbol{P}(\textbf{Blue} \mid \boldsymbol{C})}$$

Calculation:

Numerator

$$= \frac{1}{4} \times \frac{2}{15} = \frac{1}{30}$$

Denominator

$$= \frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{7}{15} + \frac{1}{4} \times \frac{2}{15}$$
$$= \frac{1}{12} + \frac{7}{30} + \frac{1}{30} = \frac{7}{20}$$
$$P(C \mid \text{Blue}) = \frac{\frac{1}{30}}{\frac{7}{20}} = \frac{2}{21} = 0.0952$$

Result:

$$\boxed{P(\text{Box C} \mid \text{Blue}) = \frac{2}{21} \approx 0.0952 = 9.52\%}$$

2. A parking facility has three sections containing cars of different colors. Section A has 5 red, 3 blue, and 2 black cars; Section B has 4 red, 6 blue, and 5 black cars; and Section C has 7 red, 2 blue, and 6 black cars. A section is chosen at random, but Section B is twice as likely to be selected as each of the other sections due to higher usage. A randomly selected car from the chosen section turns out to be blue. What is the probability that the car came from Section C?

Sol)

Given:

- Section A: 5 Red, 3 Blue, 2 Black (10 cars)
- Section B: 4 Red, 6 Blue, 5 Black (15 cars)

- Section C: 7 Red, 2 Blue, 6 Black (15 cars)

Selection probabilities:

$$P(A) = \frac{1}{4},\ P(B) = \frac{1}{2},\ P(C) = \frac{1}{4}$$

Blue probabilities:

$$P(BL \mid A) = \frac{3}{10}$$
$$P(BL \mid B) = \frac{6}{15} = \frac{2}{5}$$
$$P(BL \mid C) = \frac{2}{15}$$

Formula

$$\boldsymbol{P(C \mid BL) = \frac{P(C)P(BL \mid C)}{P(A)P(BL \mid A) + P(B)P(BL \mid B) + P(C)P(BL \mid C)}}$$

Calculation

Numerator

$$= \frac{1}{4} \times \frac{2}{15} = \frac{1}{30}$$

Denominator

$$= \frac{1}{4} \times \frac{3}{10} + \frac{1}{2} \times \frac{2}{5} + \frac{1}{4} \times \frac{2}{15}$$
$$= \frac{3}{40} + \frac{1}{5} + \frac{1}{30} = \frac{37}{120}$$
$$P(C \mid BL) = \frac{\frac{1}{30}}{\frac{37}{120}} = \frac{4}{37} = 0.1081$$

Result

$$\boxed{P(\text{Section C} \mid \text{Blue}) = \frac{4}{37} \approx 0.1081 = 10.81\%}$$

3. In a real-world medical image classification task for detecting breast cancer, a deep-learning model integrated with the K-Nearest Neighbors algorithm is applied for binaryclassification (malignant vs. benign). The dataset consists of 1800 samples per class, with 20% of the data reserved for testing. During evaluation, the model reports 210True Positives (TP), 190 True Negatives (TN), and 30 False Positives (FP). Based onthe given information, determine the missing False Negatives (FN) value, and thencalculate the Accuracy, Precision, Sensitivity (Recall), and Specificity of the classifier, explaining its effectiveness in medical diagnosis.

Sol)

Given

Total samples per class = 1800

Test data = 20%

Test samples per class

$$1800 \times 20\% = 360$$

Positive class = 360

Negative class = 360

Given:

TP = 210

TN = 190

FP = 30

FN =?

FN Calculation

$$FN = 360 - 210 = 150$$

Total test samples

$$360 + 360 = 720$$

Accuracy

$$= \frac{\boldsymbol{TP + TN}}{\boldsymbol{Total}} = \frac{210 + 190}{720} = \frac{400}{720} = 55.56\%$$

Precision

$$= \frac{\boldsymbol{TP}}{\boldsymbol{TP+FP}} = \frac{210}{210+30} = \frac{210}{240} = 87.5\%$$

Sensitivity (Recall)

$$= \frac{\boldsymbol{TP}}{\boldsymbol{TP+FN}} = \frac{210}{210+150} = \frac{210}{360} = 58.33\%$$

Specificity

$$= \frac{\boldsymbol{TN}}{\boldsymbol{TN+FP}} = \frac{190}{190+30} = \frac{190}{220} = 86.36\%$$

Result

- FN = 150
- Accuracy = 55.56%
- Precision = 87.50%
- Sensitivity = 58.33%
- Specificity = 86.36%

Effectiveness:
The model has high precision and specificity, meaning it predicts positive cases accurately and identifies healthy cases well. However, the recall is low (58.33%), so it misses many actual disease cases, which is undesirable for medical diagnosis.

9. In a spam email filtering system, an AI model classifies emails as Spam or Not Spam. Out of 400 emails (200 spam and 200 non-spam): (5 Marks)

● Spam identified as Spam: 180

● Spam identified as Not Spam: 20

● Not Spam identified as Not Spam: 170

● Not Spam identified as Spam: 30

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Sol)

Given

- Total emails = 400
- Spam emails = 200
- Non-Spam emails = 200

Confusion Matrix:

- True Positive (TP) = Spam identified as Spam = 180
- False Negative (FN) = Spam identified as Not Spam = 20
- True Negative (TN) = Not Spam identified as Not Spam = 170
- False Positive (FP) = Not Spam identified as Spam = 30

1. Accuracy

Formula:

$$\textbf{Accuracy} = \frac{\boldsymbol{TP + TN}}{\boldsymbol{TP + TN + FP + FN}} \times \mathbf{100}$$

Calculation:

$$= \frac{180 + 170}{180 + 170 + 30 + 20} \times 100$$
$$= \frac{350}{400} \times 100$$
$$= 87.5\%$$

Answer:

$$\boxed{\text{Accuracy} = 87.5\%}$$

2. Precision

Formula:

$$\textbf{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FP}} \times \mathbf{100}$$

Calculation:

$$= \frac{180}{180 + 30} \times 100$$

$$= \frac{180}{210} \times 100$$
$$= 85.71\%$$

Answer:

$$\boxed{\text{Precision} = 85.71\%}$$

3. Sensitivity (Recall)

Formula:

$$\textbf{Sensitivity} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FN}} \times \mathbf{100}$$

Calculation:

$$= \frac{180}{180 + 20} \times 100$$
$$= \frac{180}{200} \times 100$$
$$= 90.0\%$$

Answer:

$$\boxed{\text{Sensitivity} = 90.0\%}$$

4. Specificity

Formula:

$$\textbf{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN + FP}} \times \mathbf{100}$$

Calculation:

$$= \frac{170}{170 + 30} \times 100$$
$$= \frac{170}{200} \times 100$$
$$= 85.0\%$$

Answer:

$$\boxed{\text{Specificity} = 85.0\%}$$

5. F1-Score

Formula:

$$\text{F1-Score} = \frac{\mathbf{2TP}}{\mathbf{2TP + FP + FN}} \times \mathbf{100}$$

Calculation:

$$= \frac{2 \times 180}{2 \times 180 + 30 + 20} \times 100$$
$$= \frac{360}{410} \times 100$$
$$= 87.80\%$$

Answer:

$$\boxed{\text{F1-Score} = 87.80\%}$$

10. In a quality inspection system in manufacturing, an AI model identifies defective and non-defective products. Out of 500 products (250 defective and 250 non-defective): (5 Marks)

- **Defective identified as Defective: 2 30**
- **Defective identified as Non-Defective: 20**
- **Non-Defective identified as Non-Defective: 225**
- **Non-Defective identified as Defective: 25**

Compute: Accuracy, Precision, Sensitivity, Specificity, and F1-Score.

Sol)

Given

- Total products = 500
- Defective products = 250

- Non-Defective products = 250

Confusion Matrix:

- True Positive (TP) = Defective identified as Defective = 230
- False Negative (FN) = Defective identified as Non-Defective = 20
- True Negative (TN) = Non-Defective identified as Non-Defective = 225
- False Positive (FP) = Non-Defective identified as Defective = 25

1. Accuracy

Formula:

$$\textbf{Accuracy} = \frac{\boldsymbol{TP + TN}}{\boldsymbol{TP + TN + FP + FN}} \times \mathbf{100}$$

Calculation:

$$\begin{aligned} &= \frac{230 + 225}{230 + 225 + 25 + 20} \times 100 \\ &= \frac{455}{500} \times 100 \\ &= 91.0\% \end{aligned}$$

Answer:

$$\boxed{\text{Accuracy} = 91.0\%}$$

2. Precision

Formula:

$$\textbf{Precision} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FP}} \times \mathbf{100}$$

Calculation:

$$\begin{aligned} &= \frac{230}{230 + 25} \times 100 \\ &= \frac{230}{255} \times 100 \\ &= 90.20\% \end{aligned}$$

Answer:

$$\boxed{\text{Precision} = 90.20\%}$$

3. Sensitivity (Recall)

Formula:

$$\textbf{Sensitivity} = \frac{\boldsymbol{TP}}{\boldsymbol{TP + FN}} \times \mathbf{100}$$

Calculation:

$$= \frac{230}{230 + 20} \times 100$$
$$= \frac{230}{250} \times 100$$
$$= 92.0\%$$

Answer:

$$\boxed{\text{Sensitivity} = 92.0\%}$$

4. Specificity

Formula:

$$\textbf{Specificity} = \frac{\boldsymbol{TN}}{\boldsymbol{TN + FP}} \times \mathbf{100}$$

Calculation:

$$= \frac{225}{225 + 25} \times 100$$
$$= \frac{225}{250} \times 100$$
$$= 90.0\%$$

Answer:

$$\boxed{\text{Specificity} = 90.0\%}$$

5. F1-Score

Formula:

$$\textbf{F1-Score} = \frac{\boldsymbol{2TP}}{\boldsymbol{2TP + FP + FN}} \times \mathbf{100}$$

Calculation:

$$\begin{aligned}
&= \frac{2 \times 230}{2 \times 230 + 25 + 20} \times 100 \\
&= \frac{460}{505} \times 100 \\
&= 91.09\%
\end{aligned}$$

Answer:

$$\boxed{\text{F1-Score} = 91.09\%}$$